

Multimodal Topic Discovery in Web Media via von Mises-Fisher Mixture Neural Topic Models

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Abstract

Topic modeling plays a critical role in organizing and understanding large-scale web content. While neural topic models (NTMs) based on variational autoencoders (VAEs) have achieved notable success in analyzing textual data, they remain limited in addressing the multimodal nature of modern web content. Existing unimodal or multimodal extensions often suffer from posterior collapse and fail to capture the directional semantics inherent in both text and images, resulting in incoherent topics and limited interpretability. To address these challenges, we propose **MM-vNTM** (MultiModal Neural Topic Model with von Mises-Fisher Mixtures), a framework for web-scale topic discovery over multimodal data. MM-vNTM leverages pre-aligned cross-modal embeddings as inputs and jointly models document-level representations of text and image modalities in a shared hyperspherical latent space. Furthermore, it defines topics as mixtures of von Mises-Fisher (vMF) distributions in the L_2 -normalized word embedding space, explicitly capturing directional similarity. Experiments on multimedia web datasets demonstrate that MM-vNTM consistently outperforms state-of-the-art unimodal and multimodal baselines in terms of overall topic quality, highlighting its effectiveness for real-world web scenarios.

CCS Concepts

• Computing methodologies → Information extraction; Topic modeling.

Keywords

Neural Topic Modeling, Multimodal Learning, Variational Autoencoder, von Mises-Fisher distribution

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1 Introduction

Topic modeling is a widely adopted unsupervised technique for discovering latent semantic structures in text corpora [3], and most existing methods predominantly focus on unimodal textual data. With the rapid evolution of web content, however, multimodal data combining images and text have become ubiquitous across platforms such as social media. In many real-world scenarios, textual content is short, noisy, or ambiguous, which hinders coherent topic discovery, whereas accompanying images provide complementary visual semantics [6]. Integrating such visual information thus offers a promising avenue for more robust and interpretable topic modeling in real-world web scenarios.

Most existing multimodal topic models are extensions of Latent Dirichlet Allocation (LDA) [3], and therefore inherit its fundamental limitations. These include a reliance on sampling-based or variational inference, which is computationally intensive, and a dependence on word co-occurrence statistics, which thereby restricts the capacity to capture deep semantics. Furthermore, many models process textual and visual modalities independently, limiting effective cross-modal semantic integration.

To improve both inference efficiency and semantic modeling capacity, neural topic models (NTMs) based on variational autoencoders (VAEs) have emerged [15]. For instance, the Embedded Topic Model (ETM) [5] enhances topic coherence by incorporating pre-trained word embeddings, while ZeroShotTM [2] and CombinedTM [1] utilize contextualized embeddings for richer document-level representations. Despite these advances, most VAE-based NTMs adopt a Gaussian prior, which tends to concentrate the learned representations (particularly low-dimensional document-level embeddings) near the origin of the latent space, causing them to become geometrically entangled [17]. Moreover, Gaussian-based VAEs are prone to posterior collapse [4], leading to poor-quality representations and degraded topic quality.

VAEs naturally accommodate multimodal inputs through joint encoding with neural networks, and recent efforts have extended VAE-based NTMs to multimodal settings. For instance, Multimodal-ZeroShotTM [6] leverages pre-aligned image and text representations and fuses them into a shared latent space (e.g., using CLIP-style encoders [14]), while Multimodal-Contrast [6] introduces a cross-modal contrastive loss to strengthen semantic alignment across

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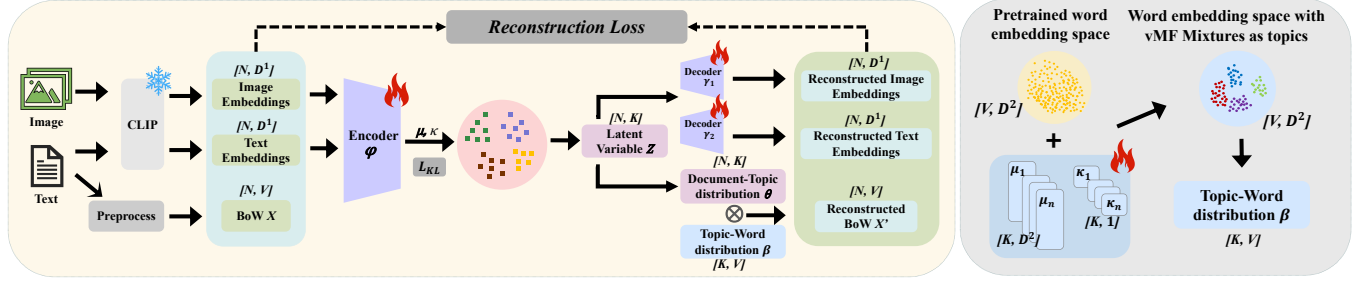


Figure 1: Overview of the model structure. The encoder ϕ takes the concatenated embeddings as input and infers the latent representation z_i for document i , while the decoders reconstruct the image and text features with z_i .

modalities. Nevertheless, these approaches still adopt a Gaussian prior and thus do not fundamentally mitigate posterior collapse, limiting the discriminability and robustness of the learned multimodal document representations. Furthermore, most existing models define topic-word relationships via matrix multiplication, where each topic is represented as a fixed point in the L_2 -normalized word embedding space. While this formulation aligns with directional similarity, it lacks explicit control over the dispersion or semantic granularity of word distributions within each topic.

The von Mises-Fisher (vMF) distribution, defined on the unit hypersphere with mean direction μ and concentration parameter κ , is well suited for modeling high-dimensional directional similarities. In vMF-based VAEs, the unit-norm constraint keeps latent codes on the sphere, avoiding collapse toward the origin and mitigating the posterior collapse common in Gaussian VAEs [4, 12]. Empirically, vMF-based models learn discriminative representations for both text and images [9, 11], and vMF-based topic models yield better-separated document embeddings, improving topic coherence and interpretability [17]. Moreover, concentration parameter κ provides principled control over semantic concentration in the word embedding space: large κ produces sharply focused topics, while small κ captures broader, multi-faceted themes. This adaptive granularity is more expressive and flexible than fixed-point topic representations for modeling heterogeneous semantics.

Motivated by these insights, we propose MM-vNTM (MultiModal Neural Topic Model with von Mises-Fisher Mixtures), a framework for topic discovery over web-scale multimodal corpora. The main contributions of this work are summarized as follows: (1) We utilize pre-aligned cross-modal embeddings as input and jointly project multimodal document representations onto a unit hyperspherical latent space. This design integrates complementary visual semantics into the topic modeling process and mitigates posterior collapse, yielding more discriminative multimodal document representations. (2) We model topics as mixtures of vMF distributions in the word embedding space, which enables adaptive control over semantic concentration and substantially improves topic interpretability and semantic coherence. (3) Extensive experiments on web-scale multimodal datasets, including quantitative comparisons, topic visualization, and ablation studies, demonstrate that MM-vNTM consistently outperforms state-of-the-art unimodal and multimodal baselines in topic quality and robustness.

2 The Proposed Method

In this section, we present MM-vNTM in detail. As illustrated in Fig. 1, the model comprises an encoder ϕ and two decoders, γ_1 and γ_2 . Let K denote the number of topics, which is the dimension of latent representation, D^1 the dimension of multimodal features, and D^2 the dimension of the word embedding space. The input consists of N paired documents $D = \{(d_i^{\text{img}}, d_i^{\text{text}})\}_{i=1}^N$. For textual representations, we preprocess and tokenize all raw text to construct a fixed vocabulary of size V . Each document is then encoded as a bag-of-words (BoW) vector, forming a matrix $X = \{x_1, \dots, x_N\} \in \mathbb{R}^{N \times V}$, where x_{iv} denotes the frequency of the v -th word in document i . For multimodal feature extraction, each image-text pair is fed into a pretrained CLIP model [14]. Its image and text encoders produce aligned embedding sets $V^{\text{img}} = \{v_1^{\text{img}}, \dots, v_N^{\text{img}}\}$ and $V^{\text{text}} = \{v_1^{\text{text}}, \dots, v_N^{\text{text}}\} \in \mathbb{R}^{N \times D^1}$. Finally, we obtain L_2 -normalized word embeddings $W = \{w_1, \dots, w_V\}$ for the vocabulary from GloVe [13].

2.1 Inference of the Latent Topic Distribution

For document i , the encoder ϕ processes the concatenated multimodal features and outputs the parameters of a vMF distribution:

$$\mu_i, \kappa_i = \phi([v_i^{\text{img}}, v_i^{\text{text}}]), \quad (1)$$

where $\mu_i \in \mathbb{R}^K$ with $\|\mu_i\| = 1$ denotes the mean direction, and $\kappa_i \in \mathbb{R}_{\geq 0}$ controls the concentration. The latent topic vector $z_i \in \mathbb{R}^K$ is then sampled from the corresponding vMF distribution:

$$z_i \sim \text{vMF}(\mu_i, \kappa_i). \quad (2)$$

To enable end-to-end training, we employ a differentiable reparameterization strategy for vMF distributions. For $K > 3$, we use rejection sampling [4] to compute a radial component r_i . For $K = 3$:

$$r_i = 1 + \frac{1}{\kappa_i} \log(u_i + (1 - u_i)e^{-2\kappa_i}), \quad u_i \sim \text{Uniform}(0, 1). \quad (3)$$

We then sample a unit vector s_i orthogonal to μ_i via normalized Gaussian projection, and construct an intermediate vector z'_i :

$$z'_i = \left[\frac{r_i}{\sqrt{1 - r_i^2}} s_i \right] \in \mathbb{R}^K, \quad (4)$$

The final latent sample is transformed using the Householder operator $h(\cdot)$, and the document-topic distribution $\theta_i \in \mathbb{S}^{K-1}$ is computed via temperature-controlled softmax:

$$z_i = h(z'_i, \mu_i), \quad \theta_i = \text{softmax}(\tau \cdot z_i). \quad (5)$$

2.2 Topic Modeling via vMF Mixtures

In this model, documents generate words through a mixture of vMF-based topics. Each topic t_j ($j = 1, \dots, K$) is modeled as a vMF distribution over the word embedding space in $\mathbb{R}^{D^2}$. Let μ_j^t and κ_j^t denote the learnable directional mean and concentration parameters of topic t_j . For a word embedding $\mathbf{w}_v \in \mathbf{W}$, its (unnormalized) likelihood under topic j is given by the vMF density f_{vMF} :

$$\beta_{\mathbf{w}_v, j} = f_{\text{vMF}}(\mu_j^t, \kappa_j^t, \mathbf{w}_v) = C_{D^2}(\kappa_j^t) \exp(\kappa_j^t (\mu_j^t)^\top \mathbf{w}_v), \quad (6)$$

where $C_{D^2}(\cdot)$ is the normalization constant ensuring the distribution integrates to one, given by:

$$C_{D^2}(\kappa) = \frac{\kappa^{D^2/2-1}}{(2\pi)^{D^2/2} I_{D^2/2-1}(\kappa)}, \quad (7)$$

and $I_{D^2}(\cdot)$ denotes the modified Bessel function of the first kind.

The full topic-word distribution for topic j is obtained via softmax over the vocabulary:

$$\beta_j = \text{softmax}([\beta_{\mathbf{w}_1, j}, \dots, \beta_{\mathbf{w}_V, j}]^\top) \in \mathbb{R}^V. \quad (8)$$

2.3 Training Objective

The overall training objective combines regularization and reconstruction terms:

$$\mathcal{L} = \mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{rec}}^{\text{BoW}} + \mathcal{L}_{\text{rec}}^{\text{text}} + \mathcal{L}_{\text{rec}}^{\text{img}}, \quad (9)$$

where $\mathcal{L}_{\text{KL}}$ regularizes the posterior toward a uniform prior (i.e., $\kappa = 0$) on the unit hypersphere, thereby promoting generalization and mitigating overfitting:

$$\mathcal{L}_{\text{KL}} = \frac{1}{N} \sum_{i=1}^N \left[\kappa_i \frac{I_{\frac{K}{2}}(\kappa_i)}{I_{\frac{K}{2}-1}(\kappa_i)} + \log C_K(\kappa_i) - \log \left(\frac{\Gamma(\frac{K}{2})}{2\pi^{\frac{K}{2}}} \right) \right], \quad (10)$$

where $\Gamma(\cdot)$ is the Gamma function.

The BoW reconstruction term $\mathcal{L}_{\text{rec}}^{\text{BoW}}$ measures the negative log-likelihood of generating document i given its inferred topic distribution. Since words may arise from multiple topics, the likelihood of each word is modeled as a mixture over topic-word distributions:

$$\mathcal{L}_{\text{rec}}^{\text{BoW}} = - \sum_{i=1}^N \sum_{v=1}^V \mathbf{x}_{iv} \log \left(\sum_{j=1}^K \theta_{ij} \cdot \beta_{jv} \right). \quad (11)$$

To ensure faithful multimodal representation learning, we employ two decoders γ_1 and γ_2 that reconstruct the original image and text features from $\mathbf{z}_i$:

$$\hat{\mathbf{v}}_i^{\text{img}} = \gamma_1(\mathbf{z}_i) \in \mathbb{R}^{D^1}, \quad \hat{\mathbf{v}}_i^{\text{text}} = \gamma_2(\mathbf{z}_i) \in \mathbb{R}^{D^1}. \quad (12)$$

For each modality $m \in \{\text{img}, \text{text}\}$, reconstruction quality is measured via a cosine embedding loss:

$$\mathcal{L}_{\text{rec}}^m = \frac{1}{N} \sum_{i=1}^N \left[1 - \frac{(\hat{\mathbf{v}}_i^m)^\top \mathbf{v}_i^m}{\|\hat{\mathbf{v}}_i^m\| \|\mathbf{v}_i^m\|} \right]. \quad (13)$$

Collectively, this objective ensures that: (1) the KL regularization term prevents posterior collapse and enforces a well-structured hyperspherical latent space; (2) the BoW term faithfully reconstructs documents through interpretable topic-word relationships grounded in the embedding geometry; and (3) the multimodal decoders preserve cross-modal semantics by reconstructing both image and text embeddings from a shared latent representation.

3 Experimental Results

We conduct experiments on three widely used image-text corpora: Twitter100k [8], MS COCO [10] and Flickr30k [18]. Our proposed MM-vNTM model is evaluated against the classical baseline LDA [3], the representative clustering-based approach BERTopic [7] and a suite of state-of-the-art VAE-based NTMs: NeuralLDA [15], ETM [5], ZeroShotTM [2], CombinedTM [1], vONT [17], NVGTM [16], Multimodal-ZeroShotTM and Multimodal-Contrast [6]. In our MM-vNTM, we initialize the concentration parameter as $\kappa = 50$ for the encoder and $\kappa = 10$ for the topics in the word embedding space.

Table 1: Topic modeling performance on different datasets under different settings of topic number (10, 20, 50).

	Num of Topics	10			20			50		
		C_v	TD	Quality	C_v	TD	Quality	C_v	TD	Quality
Twitter100k	LDA (2003)	0.370	0.860	0.318	0.310	0.985	0.305	0.271	0.974	0.264
	NeuralLDA (2017)	0.395	1.000	0.395	0.338	1.000	0.338	0.381	0.954	0.364
	ETM (2020)	0.360	0.210	0.076	0.365	0.185	0.068	0.380	0.174	0.066
	ZeroShotTM (2021)	0.401	0.950	0.381	0.412	0.675	0.278	0.448	0.470	0.211
	CombinedTM (2021)	0.403	0.930	0.375	0.432	0.675	0.292	0.463	0.522	0.241
	BERTopic (2022)	0.411	0.920	0.378	0.491	0.910	0.447	0.381	0.940	0.358
	vONT (2023)	0.409	0.970	0.397	0.330	1.000	0.330	0.337	1.000	0.337
	NVGTM (2023)	0.385	0.680	0.262	0.327	0.520	0.170	0.347	0.370	0.128
	Multimodal-ZeroShotTM (2024)	0.378	0.960	0.363	0.385	0.790	0.304	0.438	0.490	0.215
	Multimodal-Contrast (2024)	0.506	0.860	0.435	0.501	0.900	0.451	0.503	0.824	0.414
	MM-vNTM (Ours)	<u>0.508</u>	<u>1.000</u>	<u>0.508</u>	<u>0.511</u>	<u>1.000</u>	<u>0.511</u>	<u>0.580</u>	<u>0.982</u>	<u>0.569</u>
MS COCO	LDA (2003)	0.420	0.880	0.370	0.428	0.835	0.357	0.378	0.756	0.286
	NeuralLDA (2017)	0.329	1.000	0.329	0.345	<u>1.000</u>	0.345	0.371	0.822	0.305
	ETM (2020)	0.240	0.120	0.029	0.236	0.085	0.020	0.231	0.062	0.014
	ZeroShotTM (2021)	0.369	0.920	0.339	0.441	0.720	0.317	0.368	0.318	0.117
	CombinedTM (2021)	0.373	0.970	0.362	0.392	0.780	0.306	0.394	0.394	0.155
	BERTopic (2022)	0.441	0.850	0.375	0.463	0.765	0.354	0.498	0.698	0.348
	vONT (2023)	0.358	1.000	0.358	0.354	0.980	0.347	0.353	<u>0.972</u>	0.343
	NVGTM (2023)	0.249	0.720	0.180	0.283	0.555	0.157	0.299	0.352	0.105
	Multimodal-ZeroShotTM (2024)	0.324	0.980	0.317	0.418	0.780	0.326	0.388	0.366	0.142
	Multimodal-Contrast (2024)	0.427	0.780	0.333	0.422	0.905	0.382	0.425	0.862	0.366
	MM-vNTM (Ours)	<u>0.474</u>	<u>1.000</u>	<u>0.474</u>	<u>0.479</u>	<u>0.995</u>	<u>0.477</u>	<u>0.513</u>	<u>0.832</u>	<u>0.427</u>
Flickr30k	LDA (2003)	0.462	0.560	0.259	0.424	0.615	0.260	0.385	0.690	0.265
	NeuralLDA (2017)	0.329	0.990	0.326	0.287	0.950	0.272	0.315	0.690	0.217
	ETM (2020)	0.366	0.190	0.070	0.362	0.135	0.049	0.381	0.112	0.043
	ZeroShotTM (2021)	0.452	0.850	0.384	0.484	0.615	0.298	0.494	0.336	0.166
	CombinedTM (2021)	<u>0.478</u>	0.760	0.363	0.480	0.630	0.303	<u>0.498</u>	0.326	0.162
	BERTopic (2022)	0.451	0.680	0.307	<u>0.504</u>	0.710	0.358	0.492	0.736	0.362
	vONT (2023)	0.372	0.960	0.357	0.351	0.980	0.344	0.312	0.768	0.239
	NVGTM (2023)	0.285	0.690	0.197	0.317	0.495	0.157	0.303	0.262	0.079
	Multimodal-ZeroShotTM (2024)	0.455	0.870	0.395	0.451	0.700	0.316	0.441	0.344	0.152
	Multimodal-Contrast (2024)	0.342	0.720	0.246	0.442	0.795	0.351	0.430	0.684	0.294
	MM-vNTM (Ours)	0.414	<u>1.000</u>	<u>0.414</u>	0.472	<u>0.990</u>	<u>0.467</u>	0.463	<u>0.806</u>	<u>0.373</u>

*Top 10 keywords of each topic are used to compute C_v and TD.

3.1 Topic Quality Analysis

We evaluate topic modeling performance using three widely adopted metrics: Coherence Value (C_v), Topic Diversity (TD), and Quality [5]. C_v measures the semantic coherence of individual topics, while TD quantifies the proportion of unique top- N words across all topics. The Quality is defined as the product of C_v and TD, capturing the trade-off between semantic tightness and non-redundancy.

As shown in Table 1, MM-vNTM consistently achieves the highest Quality scores across all datasets and topic numbers, indicating a superior balance between coherence and diversity. Although some baselines (e.g., BERTopic, Multimodal-ZeroShotTM) occasionally achieve the best C_v or TD in isolation, none match MM-vNTM’s joint optimization of both criteria. This finding demonstrates that MM-vNTM produces topics that are not only semantically coherent but also well differentiated, a crucial advantage in multimodal topic modeling where avoiding topic homogenization and covering diverse semantic aspects grounded in image-text alignment is essential. To further illustrate the semantic coherence of the discovered topics, we present selected examples from Twitter100k in Fig. 2, each consisting of keywords and the original image-text pairs of the documents with the highest topic assignment probabilities.



Figure 2: Sample topics discovered from Twitter100k (top 10 words and top-5 image-text documents by topic probability).

3.2 Ablation Study

To evaluate the contribution of individual components: hyperspherical prior of latent space (**e**), vMF mixture topic modeling in word embedding space (**d**), and image-features input (**i**), we conduct an ablation study on three variants. As shown in Fig. 3, removing any component degrades performance, confirming their complementary roles. Removing both hyperspherical prior and vMF topic modeling (**w/o e,d**) yields the lowest scores, highlighting the limitations of Gaussian priors and fixed-point topic representations. Restoring the hyperspherical encoder alone (**w/o d**) brings marginal improvement, whereas reintroducing vMF mixture modeling (**w/o i**) significantly boosts coherence. Notably, even without image input (**w/o i**), the model substantially outperforms the fully ablated variant, attesting to the efficacy of geometric topic modeling with vMF distributions. Finally, the full MM-vNTM incorporating all components achieves consistent improvements (most pronounced at 50topics), validating that visual semantics enriches document representation and mitigates text sparsity in multimodal corpora.

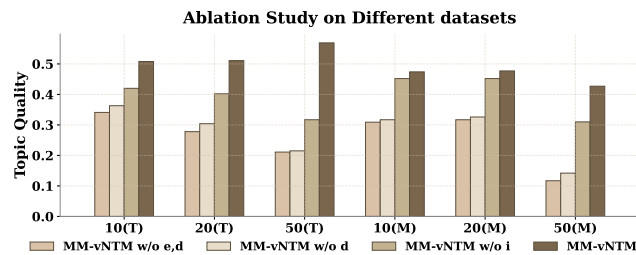


Figure 3: Ablation study results under different topic number settings (10, 20, 50) on Twitter100k (T) and MS COCO (M).

4 Conclusion

In this work, we proposed MM-vNTM, a multimodal VAE-based NTM that unifies image-text features for learning document representations and models topics as vMF mixtures over the word embedding space. By enabling adaptive control of topic concentration and jointly reconstructing multimodal representations, MM-vNTM yields more coherent, diverse, and semantically grounded topics, advancing multimodal topic modeling for real-world web data. In future work, we plan to extend MM-vNTM to additional modalities such as audio and video, and to develop online or incremental inference schemes for streaming web content.

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